



FINAL REPORT WORKSHOP: PYTHON.

**TOPIC: "DEVELOPMENT OF A MACHINE LEARNING MODEL
USING THE PYTHON PROGRAMMING LANGUAGE"**

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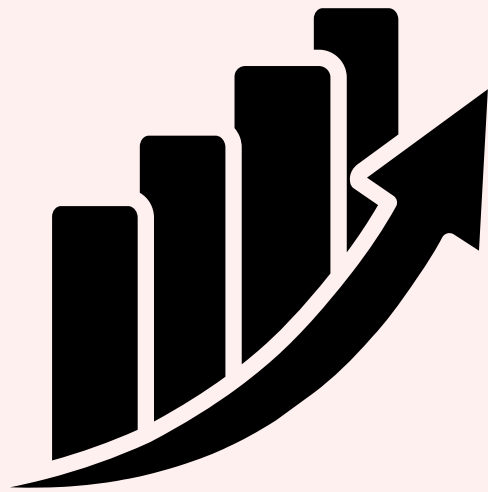
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YEAR: 2024 - 2025.



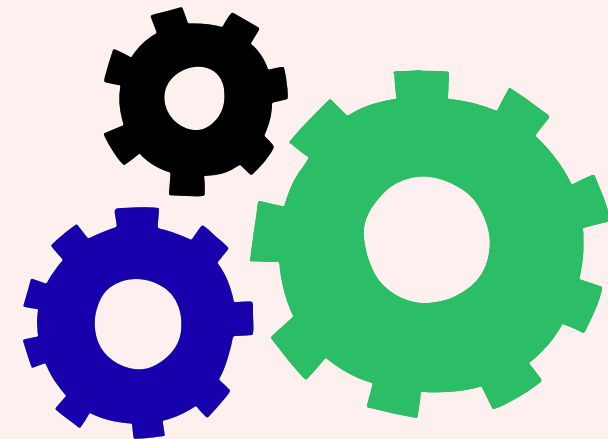
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**CONTEXTE AND
OBJECTIF**



**COMMERCIAL
REPORT**

**TECHNICAL
REPORT**



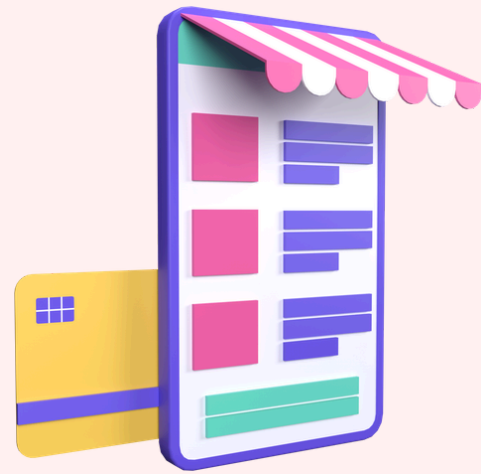
CONCLUSION

Context & OBjectif



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The dataset was designed to predict whether individuals will convert (i.e., make a purchase) based on their age, sex, interactions with the marketing company's communication channels, and other factors. Consequently, 'conversion' is the target variable. Machine learning models will be applied to make the predictions, using the Python programming language.

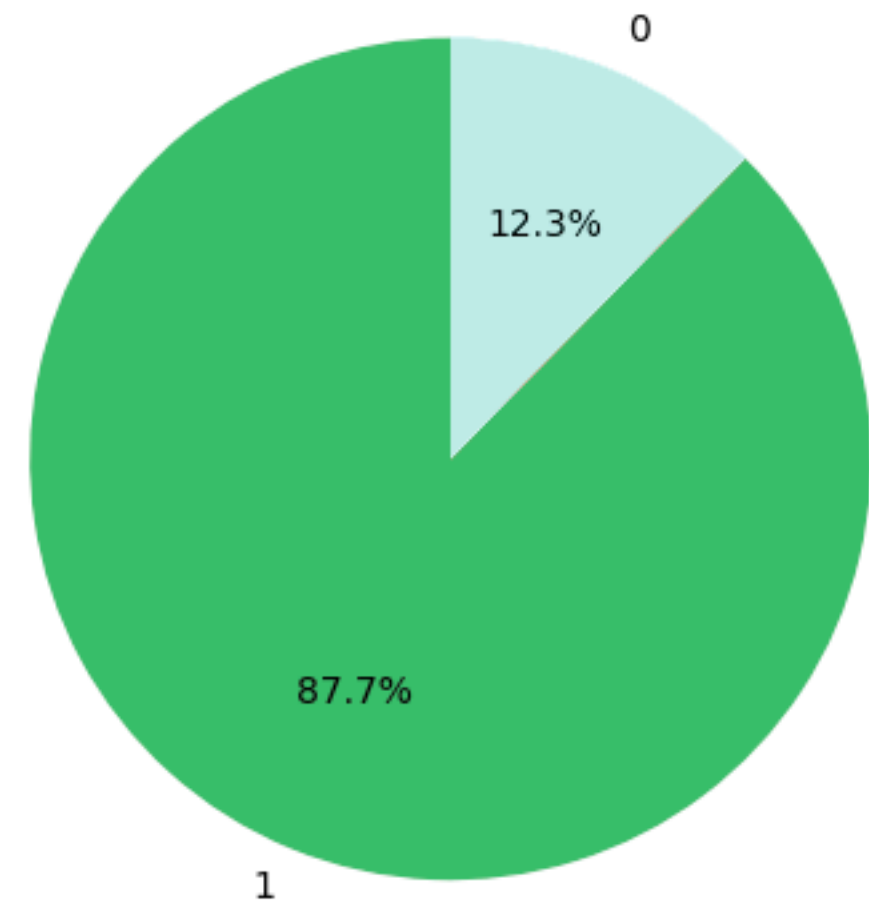


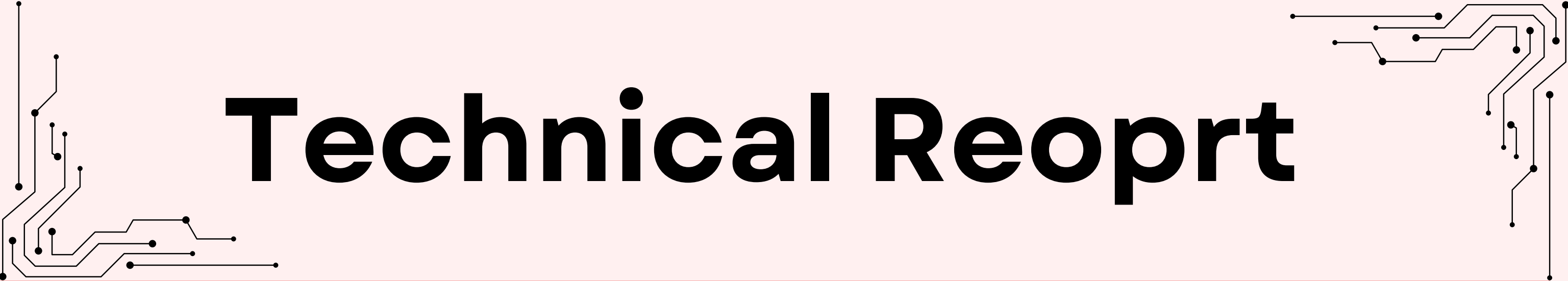
Commercial Report

Key Takeaways

- **Performance:** High conversion rate (87.7%) signals strong overall profitability and campaign efficiency.
- **Audience:** Targets adults aged 18–70 (\$20k–\$150k income). The predominantly female user base suggests sticking to a niche focus or expanding into male offerings to scale.
- **Marketing:** Balanced acquisition channels led by Referral (21.5%) and Conversion campaigns (26%). Ad spend caps at \$10k, with CTRs between 0%–30%.
- **Core Drivers:** Conversions are heavily driven by ad budget, site/email engagement, and customer loyalty (repeat buyers with high loyalty points). Retention incentives remain critical to spur organic word-of-mouth.

Répartition de la variable 'conversion'





Technical Reoprt

- **TARGETING HIGH-VALUE CONVERTERS (CLASS 1): RANDOM FOREST IS THE TOP MODEL FOR OPTIMIZING AD BUDGET AND COMMERCIAL RESOURCES. IT ACHIEVES 0.92 PRECISION AND 0.98 RECALL ($F_1 = 0.95$), CAPTURING NEARLY ALL TRUE CONVERTERS WHILE MAINTAINING HIGH TARGETING RELIABILITY TO PREVENT WASTED SPEND.**
- **DETECTING AT-RISK CUSTOMERS (CLASS 0): LOGISTIC REGRESSION IS THE BEST CHOICE FOR RETENTION AND CHURN PREVENTION. BY PRIORITIZING COVERAGE OVER PRECISION, IT CAPTURES 69% OF AT-RISK CLIENTS (RECALL). ITS LOWER PRECISION (0.30) IS AN ACCEPTABLE TRADEOFF IN RETARGETING SCENARIOS, AS MISSING AT-RISK CUSTOMERS CARRIES A HIGHER COST THAN SENDING LOW-COST FOLLOW-UP CAMPAIGNS TO NON-CHURNERS.**



Conclusion



No single model fits all operational needs:

- Random Forest is ideal for driving high-value conversions (Class 1) and maximizing marketing ROI, thanks to its high precision ($F_1 = 0.95$).
- Logistic Regression is best suited for churn prevention (Class 0), prioritizing customer reach (0.69 Recall) over precision.

Deploying a dual-model strategy ensures both efficient customer acquisition and effective retention.

**THANK
YOU FOR
YOUR
ATTENTION**

